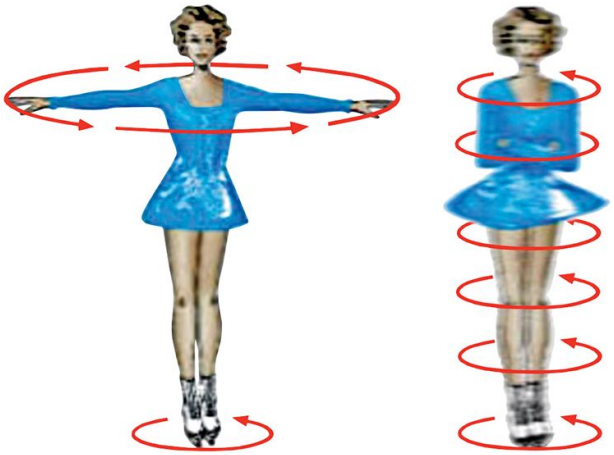


Fig. 27.4. Ice skater.

The *Endurance* heads toward Gargantua with a certain amount of energy, which like its angular momentum remains constant along its trajectory. This energy consists of three parts: the *Endurance*'s *gravitational energy*, which gets more and more negative as the *Endurance* plunges toward Gargantua; its *centrifugal energy* (its energy of circumferential motion around Gargantua), which increases as the *Endurance* plunges because the circumferential motion is speeding up; and its *radial kinetic energy* (its energy of motion toward Gargantua).

The surface in Figure 27.3 is the *Endurance*'s gravitational energy plus its centrifugal energy plotted vertically, and location in Gargantua's equatorial plane plotted horizontally. Wherever the surface dips downward, the *Endurance*'s gravitational plus centrifugal energy decreases, so its radial kinetic energy must increase (since the total energy is unchanged); its radial motion must speed up. This is precisely what happens in our intuitive, volcano analogy.

Outside the moat of Figure 27.3, the surface's height is controlled by the *Endurance*'s negative gravitational energy (see the "gravitational energy" label on the figure). By comparison, there the positive centrifugal energy is unimportant. On the outer edge of the volcano, by contrast, the height is controlled by the rising centrifugal energy, which has come to dominate over the gravitational energy. On the inside of the